

Motivation: broad hidden-training detection needs independent low-cost evidence

Method and measured application scope

Prior NVML

162 workloads (106 training / 40 inference / 16 other)

Unseen family: 0/23 detected; 10/95 false positives

WAVE

3 decoder families: GPT-2, LLaMA, Qwen

Architecture verification only; not training detection

SensorGuard

Targets: ResNet-50, GPT-2, BERT (+ MLP in the physical pilot)

Controls: 4 inference + 8 non-ML; fused update pending

Measured monitoring overhead

continuous budget $\leq 1\%$

0.0% in our logger check

0.0% base logger*

*physical logger pending

2,750% mean

range 2,254-3,059%

0 1 10 100 1000

Workload penalty (%) - symlog

Roofline ambiguity: 101/286 non-training configurations fell inside the training arithmetic-intensity range - characterization is not detection.
SensorGuard's end-to-end physical-logger overhead and matched held-out-family result remain pending.